

PROBLEMS ON ASSOCIATED FORMS, VERONESE GEOMETRY, AND LEFSCHETZ PROPERTIES

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ABSTRACT. We discuss three problems connecting Lefschetz properties, Macaulay inverse systems, higher Hessians, and Veronese geometry. The first concerns higher Hessians of the associated form of an Artinian complete intersection. The second asks whether higher Lefschetz maps can be understood through osculating spaces of Veronese varieties. The third comes from singular Veronese-Avoiding hypersurfaces and asks for a better understanding of the case of fewer than n nodes. We recall the main results that lead to these questions and describe some concrete cases that may serve as starting points.

1. INTRODUCTION

Let $S = \mathbb{k}[x_1, \dots, x_n]$ be a standard graded polynomial ring over an algebraically closed field $\mathbb{k}$ of characteristic zero. For $m \geq 1$, let

$$V_n^m = \{[\ell^m] \mid 0 \neq \ell \in S_1\} \subseteq \mathbb{P}(S_m)$$

denote the degree- m Veronese variety. Let

$$J = (f_1, \dots, f_n) \subseteq S, \quad \deg(f_i) = d_i \geq 2,$$

be a homogeneous Artinian complete intersection and set $A = S/J$. Then A is Artinian Gorenstein with socle degree

$$T = \sum_{i=1}^n (d_i - 1).$$

A basic open problem asks whether A always has the Strong Lefschetz Property. Since A is Gorenstein, this is equivalent to asking whether, for a general linear form $\ell \in A_1$,

$$\times \ell^{T-2k} : A_k \longrightarrow A_{T-k}$$

is an isomorphism for every $0 \leq k < T/2$. We write SLP_k for the corresponding property in degree k . Monomial complete intersections have the Strong Lefschetz Property in characteristic zero, and height-three complete intersections have the Weak Lefschetz Property, but the Strong Lefschetz Property is open in general; see [12, 14, 7, 6].

Macaulay duality gives another way to look at this question. If F is a Macaulay inverse system for A , then the Lefschetz maps are controlled by the higher Hessians of F ; see [11]. For a complete intersection, this dual generator is usually called the *associated form*. Thus one can study the complete-intersection Lefschetz problem through the geometry of one homogeneous polynomial.

A second point of view comes from Veronese varieties. In [4], the degree-one Lefschetz map was related to the position of $\mathbb{P}(J_{T-1})$ with respect to the Veronese variety of $(T-1)$ -st powers of linear forms. This suggests that higher Lefschetz maps should be related to higher osculating spaces of Veronese varieties.

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The same geometry survives, in a different form, for singular hypersurfaces. In [9], Veronese-Avoiding hypersurfaces were introduced by imposing a condition on the degree- $(T-1)$ part of the gradient ideal. If a hypersurface has exactly n isolated singular points, the condition has a simple geometric meaning: the singularities must be n nodes in general linear position. With fewer than n nodes, however, the situation changes. The answer is no longer determined by the local singularities and is governed by a natural rational map. This gives the third problem below.

These three problems are closely related. The first is algebraic, the second asks for a geometric interpretation of the same phenomenon, and the third asks how much of this picture remains in the singular setting.

2. ASSOCIATED FORMS AND HIGHER HESSIANS

Let

$$S^\vee = \mathbb{k}[y_1, \dots, y_n],$$

and let $S = \mathbb{k}[x_1, \dots, x_n]$ act on $S^\vee$ by differentiation, with x_i acting as $\partial/\partial y_i$. For $F \in S_T^\vee$, set

$$\text{Ann}(F) = \{\alpha \in S \mid \alpha \circ F = 0\}.$$

Macaulay duality says that every standard graded Artinian Gorenstein algebra of socle degree T is of the form $S/\text{Ann}(F)$ for a homogeneous form $F \in S_T^\vee$, unique up to a nonzero scalar; see [10, 8].

For the complete intersection $J = (f_1, \dots, f_n)$ fixed above, we write $A_{\mathbf{f}}$ for its associated form, so that

$$J = \text{Ann}(A_{\mathbf{f}}).$$

Set $h_k = \dim_{\mathbb{k}} A_k$, and let $\mathcal{B}_k = \{\alpha_1, \dots, \alpha_{h_k}\}$ be a basis of A_k . The k -th Hessian of $A_{\mathbf{f}}$ is, up to a nonzero scalar,

$$\text{hess}_{A_{\mathbf{f}}}^{(k)} = \det(\alpha_i \alpha_j \circ A_{\mathbf{f}})_{i,j}.$$

For $k = 1$ this is the ordinary Hessian.

The theorem of Maeno–Watanabe gives the precise link with Lefschetz maps.

Theorem 2.1 (Maeno–Watanabe [11]). *Let $A = S/\text{Ann}(F)$ be Artinian Gorenstein of socle degree T . For $0 \leq k \leq \lfloor T/2 \rfloor$, the multiplication map*

$$\times L^{T-2k} : A_k \longrightarrow A_{T-k}$$

is an isomorphism for a general linear form L if and only if $\text{hess}_F^{(k)}$ is not the zero polynomial.

This leads to our first problem.

Problem 1 (Higher Hessians of associated forms). Let $A_{\mathbf{f}}$ be the associated form of a homogeneous Artinian complete intersection. Must

$$\text{hess}_{A_{\mathbf{f}}}^{(k)} \neq 0$$

for every $1 \leq k < T/2$?

By the theorem above, this is exactly the Strong Lefschetz problem for homogeneous Artinian complete intersections. It is useful to separate what is already known from what is still open.

The ordinary Hessian. The first case is $k = 1$. Since $\text{Ann}(A_{\mathbf{f}})_1 = 0$, the associated form is not a cone. The classical theorem of Gordan and Noether says that a non-conical form in at most four variables cannot have identically vanishing Hessian. In the language of Artinian Gorenstein algebras, SLP_1 therefore holds in codimension at most four; see [3].

The first new embedding dimension is thus $n = 5$. There are non-conical forms in five variables with zero Hessian, but it is not known whether one of them can occur as the associated form of a complete intersection. There is one important positive result: a complete intersection of five quadrics has the full Strong Lefschetz Property [3]. Thus natural first cases include

$$(2, 2, 2, 2, 3), \quad (2, 2, 2, 3, 3), \quad (3, 3, 3, 3, 3).$$

The last case already has a familiar geometric source. If $g \in \mathbb{k}[x_1, \dots, x_5]$ defines a smooth quartic threefold in $\mathbb{P}^4$, then its gradient ideal is a complete intersection of five cubics. Hence the multidegree $(3, 3, 3, 3, 3)$ contains the question whether the associated form of the Milnor algebra of a smooth quartic threefold can have vanishing Hessian.

There is also a useful generic result. For every fixed multidegree, [4] proves that a generic complete intersection satisfies

$$\mathbb{P}(J_{T-1}) \cap V_n^{T-1} = \emptyset,$$

and hence has SLP_1 . In the same setting, this avoidance condition is equivalent to the smoothness of the hypersurface defined by the associated form. Thus, if a counterexample to SLP_1 exists, it must lie in a special locus of the space of complete intersections. The degree-one Hessian criterion is the case $k = 1$ of the Maeno–Watanabe theorem above; see also Wang [13] for a recent treatment in the complete-intersection setting.

Higher Hessians. For arbitrary Artinian Gorenstein algebras, higher Hessians can vanish [5]. Therefore Problem 1 is really about the extra structure forced by a complete-intersection apolar ideal. The question is whether this extra structure prevents the higher Lefschetz maps from degenerating.

There are two natural places to begin. One is the ternary case. Height-three complete intersections have the Weak Lefschetz Property, but this does not settle the higher maps

$$\times \ell^{T-2k} : A_k \longrightarrow A_{T-k}.$$

The first nontrivial higher Hessians in small degree can therefore be studied explicitly. The other is codimension five, where the ordinary Hessian problem is already genuinely new.

A useful way to think about Problem 1 is this: among all Artinian Gorenstein algebras, vanishing higher Hessians are possible, but we do not know whether the locus of such algebras meets the complete-intersection locus. This makes the geometry of the associated-form construction itself part of the problem.

3. OSCULATING SPACES AND HIGHER LEFSCHETZ MAPS

The degree-one argument in [4] uses tangent spaces to a Veronese variety. There is a natural higher-order version, and it gives a direct geometric form of the Lefschetz maps.

Let $A = S/J$ be Artinian Gorenstein with socle degree T , fix $1 \leq k < T/2$, and let $0 \neq \ell \in S_1$. The kernel of

$$\times \ell^{T-2k} : A_k \longrightarrow A_{T-k}$$

is

$$\frac{(J : \ell^{T-2k})_k}{J_k}.$$

Hence the map is an isomorphism if and only if

$$(1) \quad J_{T-k} \cap \ell^{T-2k} S_k = \ell^{T-2k} J_k.$$

Now consider the Veronese embedding

$$\nu_{T-k} : \mathbb{P}(S_1) \longrightarrow \mathbb{P}(S_{T-k}), \quad [\ell] \longmapsto [\ell^{T-k}].$$

Its k -th osculating space at $[\ell^{T-k}]$ is

$$\mathrm{Osc}_{[\ell^{T-k}]}^k(V_n^{T-k}) = \mathbb{P}(\ell^{T-2k}S_k);$$

see, for instance, [2]. Therefore, when $J_k = 0$, condition (1) becomes the literal avoidance condition

$$\mathbb{P}(J_{T-k}) \cap \mathrm{Osc}_{[\ell^{T-k}]}^k(V_n^{T-k}) = \emptyset.$$

For a complete intersection of multidegree $d_1 \leq \dots \leq d_n$, this applies whenever $k < d_1$.

When $J_k \neq 0$, the intersection is not expected to be empty. The correct quantity is the excess intersection

$$E_k(\ell) := \frac{J_{T-k} \cap \ell^{T-2k}S_k}{\ell^{T-2k}J_k}.$$

Multiplication by ℓ^{T-2k} induces a canonical isomorphism from the kernel of the Lefschetz map onto this vector space. Thus

$$E_k(\ell) = 0$$

for a general ℓ if and only if SLP_k holds.

This also gives a natural degeneracy locus

$$\mathcal{D}_k(J) = \{[\ell] \in \mathbb{P}(S_1) \mid E_k(\ell) \neq 0\}.$$

After choosing bases, the determinant of the multiplication map is, up to a nonzero scalar, the k -th Hessian of the Macaulay dual generator evaluated at the coefficients of ℓ ; this is another form of the Maeno–Watanabe criterion. Thus higher Hessians and osculating spaces describe the same failure from two different sides.

Problem 2 (Osculating spaces and Lefschetz maps). Let J be a homogeneous Artinian complete intersection of socle degree T , and fix $1 \leq k < T/2$. For a general linear form ℓ , understand the excess space

$$E_k(\ell) = \frac{J_{T-k} \cap \ell^{T-2k}S_k}{\ell^{T-2k}J_k}.$$

Can the complete-intersection structure of J be used to prove $E_k(\ell) = 0$, or equivalently,

$$J_{T-k} \cap \ell^{T-2k}S_k = \ell^{T-2k}J_k?$$

The cleanest range is $k < d_1$, where $J_k = 0$. Then the problem asks whether

$$\mathbb{P}(J_{T-k}) \cap \mathrm{Osc}_{[\ell^{T-k}]}^k(V_n^{T-k}) = \emptyset$$

for general ℓ . For $k = 1$ this is the tangent-space picture used in [4]. The first higher-order case is $k = 2$; when $d_1 \geq 3$, it already lies in this clean range. Even in three variables this seems to be a useful test case: the Veronese osculating spaces are explicit, while the complete-intersection condition keeps strong control over the Hilbert function of S/J .

There are several possible ways to approach the problem. In the clean range $k < d_1$, one can study the incidence correspondence of pairs $([\ell], [F])$ with

$$[F] \in \mathbb{P}(J_{T-k}) \cap \mathrm{Osc}_{[\ell^{T-k}]}^k(V_n^{T-k}).$$

For general k , the degeneracy locus $\mathcal{D}_k(J)$ is the more natural object. It would also be interesting to understand whether the Gorenstein approach to the Gordan–Noether theorem in [3], which is effective for $k = 1$, has a useful higher-order analogue. A geometric explanation for the expected nonvanishing would be especially useful.

4. SINGULAR VERONESE-AVOIDING HYPERSURFACES

We now turn to singular hypersurfaces. From this point on, associated forms are not needed. The relevant objects are the gradient ideal, the Milnor algebra, and the Jacobian module.

Let

$$R = \mathbb{k}[x_1, \dots, x_n], \quad f \in R_d,$$

with f reduced and $d \geq 3$. Set $X = V(f) \subseteq \mathbb{P}^{n-1}$,

$$J_f = (f_{x_1}, \dots, f_{x_n}), \quad M_f = R/J_f, \quad T = n(d-2).$$

Following [9], X is called *Veronese-Avoiding* if

$$\dim_{\mathbb{k}}(M_f)_{T-1} = n$$

and

$$\mathbb{P}((J_f)_{T-1}) \cap V_n^{T-1} = \emptyset.$$

When X is smooth, this condition is equivalent to the smoothness of the projective hypersurface defined by the associated form, or equivalently to the nonvanishing of its discriminant; see [1, 9]. More generally, if X is Veronese-Avoiding, then

$$\times \ell^{T-2} : (M_f)_1 \longrightarrow (M_f)_{T-1}$$

is an isomorphism for a general linear form ℓ .

For singular hypersurfaces there is a sharp result when the number of singular points is exactly n .

Theorem 4.1 ([9]). *Let $X = V(f) \subseteq \mathbb{P}^{n-1}$ be reduced and suppose that X has exactly n isolated singular points. Then X is Veronese-Avoiding if and only if these singularities are ordinary nodes in general linear position.*

The case of fewer than n nodes is different. Suppose

$$\Gamma = \text{Sing}(X) = \{p_1, \dots, p_r\}, \quad r < n,$$

consists of nodes imposing independent linear conditions on R_1 . Put $I = I(\Gamma)$, $m = T - 1$, and

$$N(f) = J_f^{\text{sat}}/J_f.$$

Then the numerical part of the Veronese-Avoiding condition is automatic, and the graded self-duality of the Jacobian module gives

$$\dim_{\mathbb{k}} I_1 = \dim_{\mathbb{k}} N(f)_m = n - r.$$

Moreover, [9] shows that X is Veronese-Avoiding if and only if the rational map

$$\phi_f : \mathbb{P}(I_1) \dashrightarrow \mathbb{P}(N(f)_m), \quad [\ell] \longmapsto [\ell^m \bmod (J_f)_m],$$

has no base points. Thus, after choosing bases,

$$\phi_f : \mathbb{P}^{n-r-1} \dashrightarrow \mathbb{P}^{n-r-1}.$$

This criterion makes clear why the case $r < n$ is global. The singular points alone do not determine the degree- m piece of the Jacobian ideal. A simple family already shows the difference. For $d \geq 4$, the plane curves

$$f_d = xyz^{d-2} + x^d + y^d$$

and

$$g_d = xyz^{d-2} + x^d + y^d + x^{d-1}z$$

have the same unique singular point $[0 : 0 : 1]$, and in both cases this point is a node. Nevertheless, $V(f_d)$ is not Veronese-Avoiding, whereas $V(g_d)$ is Veronese-Avoiding; see [9]. Thus no criterion using only the number and local analytic type of the singularities can solve the problem.

Problem 3 (The $r < n$ nodal case). Fix n, d and $1 \leq r < n$. On the stratum of degree- d hypersurfaces whose singular locus consists of r nodes imposing independent linear conditions, understand the base points of

$$\phi_f : \mathbb{P}^{n-r-1} \dashrightarrow \mathbb{P}^{n-r-1}.$$

In particular, decide when this map is everywhere defined and describe the exceptional locus where it acquires a base point.

There are several closely related questions around Problem 3. A first one is whether the Veronese-Avoiding locus is nonempty on a given irreducible r -nodal stratum. Since the avoidance condition is open on the rank stratum where the numerical condition holds, one example would imply density in the irreducible component containing that example. It is also natural to ask for equations for the complement. The map ϕ_f suggests looking for such equations through resultants or other invariants of the Jacobian module.

Two cases are especially concrete. If $r = n - 1$, there is a unique hyperplane through the nodes. If H is its equation, then X is Veronese-Avoiding exactly when

$$H^{T-1} \notin (J_f)_{T-1}.$$

Thus the whole problem is reduced to one membership question in the top Jacobian piece.

If $r = n - 2$, then

$$\phi_f : \mathbb{P}^1 \dashrightarrow \mathbb{P}^1.$$

Choose coordinates and write

$$\phi_f([s : t]) = [P_f(s, t) : Q_f(s, t)],$$

where P_f and Q_f are binary forms of degree m . Then

$$X \text{ is Veronese-Avoiding} \iff \text{Res}(P_f, Q_f) \neq 0.$$

When the resultant is nonzero, ϕ_f is a morphism of degree $m = T - 1$; the interesting questions are therefore to understand the resultant itself, its vanishing locus, and the ramification of ϕ_f . For a plane curve with one node, $n = 3$ and $r = 1$, so this is exactly the situation above, with $m = 3d - 7$.

There is also a parameter-space version of the same problem. The Veronese-Avoiding locus $\mathcal{V}_{n,d} \subseteq \mathbb{P}(R_d)$ is locally closed, and [9] constructs inside it a nonempty irreducible constructible locus of hypersurfaces with exactly n nodes in general linear position. For $r < n$, the map ϕ_f suggests a finer stratification by the size and geometry of its base locus. Understanding these strata is a natural global version of Problem 3.

The three problems fit together naturally. Problem 1 asks whether complete intersections can have vanishing higher Hessians. Problem 2 asks for a geometric reason for the expected nonvanishing. Problem 3 asks how much of this Veronese geometry survives when the Milnor algebra is no longer Artinian. We hope that the special cases above provide useful starting points for the problem session.

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